

TANISHK GANGWAR

B.Tech CSE (Data Science) · Manipal University Jaipur · Batch 2028

tanishk7531@gmail.com | +91-6232509501 | github.com/tanistheta | linkedin.com/in/tanishk-gangwar | tanishkgangwar.vercel.app

EDUCATION

B.Tech — Computer Science & Engineering (Data Science), Manipal University Jaipur Oct 2024 – Present
CGPA: 9.00 / 10 — Dean's Academic Excellence Award, three consecutive semesters (Sem 2, 3 & 4)
Coursework: Data Structures & Algorithms · Machine Learning · Data Mining · DBMS · OOP · Computer Networks · Cloud Computing

TECHNICAL SKILLS

Languages	Python, C++, Java, SQL
ML / Data	Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering, Data Visualization
Statistics	Causal Inference (Difference-in-Differences), Hypothesis Testing, Cross-Validation, Information Theory
Web / APIs	FastAPI, Flask, REST APIs, React, GitHub API, JSON
Tools	Git, GitHub, Jupyter Notebook, VS Code

PROJECTS

Sekivara — Causal Analysis of GitHub Copilot's Behavioral Fingerprint  [GitHub](#) Python · Pandas · DiD

- Designed a quasi-experimental **difference-in-differences study** measuring how GitHub Copilot adoption changed developer behavior, analyzing **~403,000 commits** across 9 open-source repositories with treated/control assignment and parallel-trends validation.
- Found treated repositories shifted toward **smaller, more atomic commits** post-adoption; decomposed the effect by contributor cohort to show it was driven by **existing contributors changing behavior**, not an influx of new ones.
- Built a fully reproducible pipeline from data collection to statistical analysis — centralized configuration, automated repo ingestion via the GitHub API, and documented methodology targeting **MSR-grade rigor**.

Entropic Fingerprint — Shannon Entropy Release Detection (Negative Result)  [GitHub](#) Python · Scikit-learn

- Investigated whether **Shannon entropy** computed over file-level commit churn can detect software release events, building a feature pipeline (commit cadence, entropy delta, author diversity) over 6 years of history from Flask, Django, and CPython (2018–2024).
- Exposed a validation flaw in the initial 72%-accuracy result through proper **cross-repository cross-validation**; the corrected model scored **AUC 0.47**, establishing that entropy carries no independent release signal — documented and published as a negative result.
- Diagnosed the underlying confound — **Spearman $r = 0.817$** between commit volume and entropy — showing entropy largely proxies raw activity; open-sourced full methodology and replication code.

ACHIEVEMENTS & AWARDS

- **Dean's Academic Excellence Award** — three consecutive semesters (Sem 2, 3 & 4); CGPA 9.00/10.
- **Google — The Big Code** — ranked in the **Top 1,500 nationally** in Google's competitive programming event.
- **Smart India Hackathon (SIH) 2025** — College Winner; selected among the winning teams at the college level.

CERTIFICATIONS

- **Exploratory Data Analysis for Machine Learning** — IBM / Coursera · Credential ID: YGOFNC9BU8JD
- **The Nuts and Bolts of Machine Learning** — Google / Coursera · Credential ID: O69XDYVQNQO9